Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) One mm over one km^2 equals…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. 1 m^3

B. 1 million m^3

C. 1000 liters

D. 1000 m^3

Correct: D

2. (2 points) Dissolved load consists of…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only gravel

B. Only floating wood

C. Only sand

D. Ions and molecules in solution

Correct: D

3. (2 points) High porosity always means high permeability?

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Always

B. Only for clay

C. Only below water table

D. No

Correct: D

4. (2 points) A budget requires…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Only a river name

B. Only a map color

C. Only a starting temperature

D. A boundary and interval

Correct: D

5. (2 points) Competence describes…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Total water volume

B. Only sediment color

C. Only rainfall

D. Largest transportable particles

Correct: D

6. (2 points) Darcy flux equals…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Q x A

B. K/A

C. Porosity alone

D. Q/A

Correct: D

7. (2 points) Infiltrated water can later…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Never move again

B. Only evaporate instantly

C. Create salt by definition

D. Support streamflow

Correct: D

8. (2 points) Equal discharge guarantees equal velocity?

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Always

B. Only for rivers

C. Only at flood stage

D. No

Correct: D

9. (2 points) Flow generally follows…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Only surface slope

B. Increasing head always

C. Only north

D. Decreasing hydraulic head

Correct: D

10. (4 points) What is missing from a simple P-Q budget?

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

answer: Evapotranspiration, storage change, and other boundary transfers may matter. (4 points).

11. (2 points) A freshwater catchment receives 110 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.

answer: 40 mm (2 points).

Numerical target 40; absolute tolerance 0.08. Accepted units: , mm.

12. (2 points) A rectangular stream is 5 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.

answer: 5 m^3/s (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , m^3/s.

13. (2 points) A river channel is 13 km long between points 9 km apart along the valley. Find sinuosity.

answer: 1.4444444444444444 (2 points).

Numerical target 1.4444444; absolute tolerance 0.0028888889. Accepted units: , .

14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.

answer: 0.02 (2 points).

Numerical target 0.02; absolute tolerance 4e-05. Accepted units: , .

15. (2 points) Use Q=K A i with K=0.001 m/s, A=21 m^2 and i=0.02. Find saturated Darcy discharge.

answer: 0.00042 m^3/s (2 points).

Numerical target 0.00042; absolute tolerance 8.4e-07. Accepted units: , m^3/s.

16. (2 points) A lake stores 1000 m^3 and exports 100 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.

answer: 10 days (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , days.

17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.

answer: Faster runoff routing can shorten lag time and increase peak discharge; drainage infrastructure, storage and storm properties must be controlled. (2 points).

18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.

answer: Erosion is commonly concentrated on the outer cut bank and deposition on the inner point bar, with local flow conditions controlling details. (2 points).

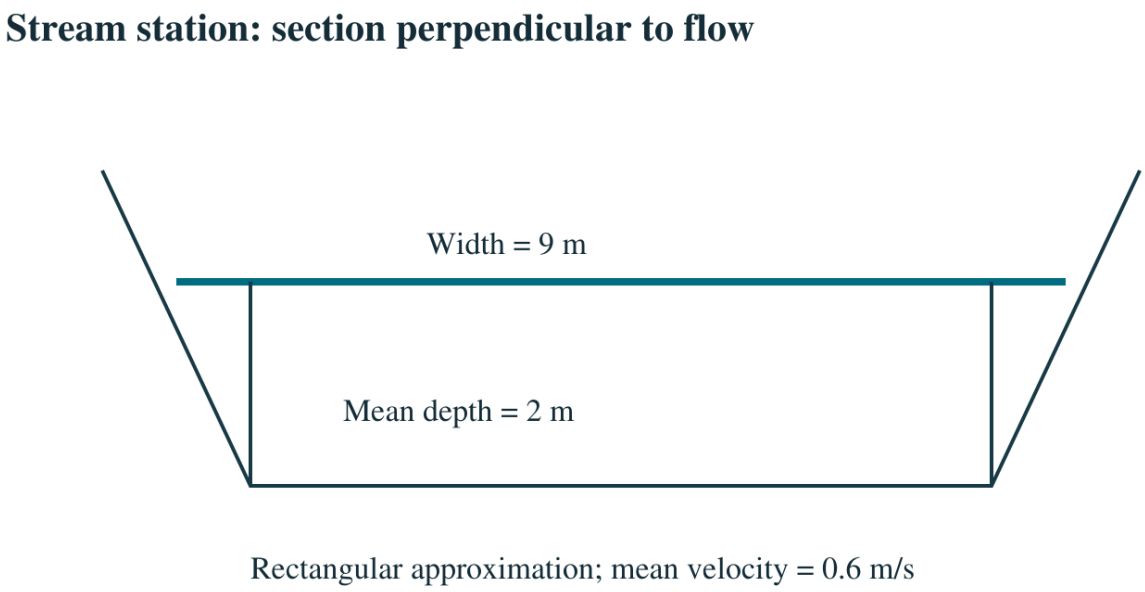
19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?

answer: Second order; joining two channels of the same order increases order by one. (2 points).

20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.

answer: Lower groundwater head can reduce discharge to the river or induce infiltration from it, depending on hydraulic connection and gradients. (2 points).

21. (2 points) Use Figure F1. Estimate wetted cross-sectional area using the indicated rectangular approximation.



F1. Stream cross section and measured velocity (original synthetic teaching figure)

answer: 18 m^2 (2 points).

Numerical target 18; absolute tolerance 0.054. Accepted units: , m^2.

22. (2 points) Use Figure F1. Calculate stream discharge from this section.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

answer: 10.8 m^3/s (2 points).

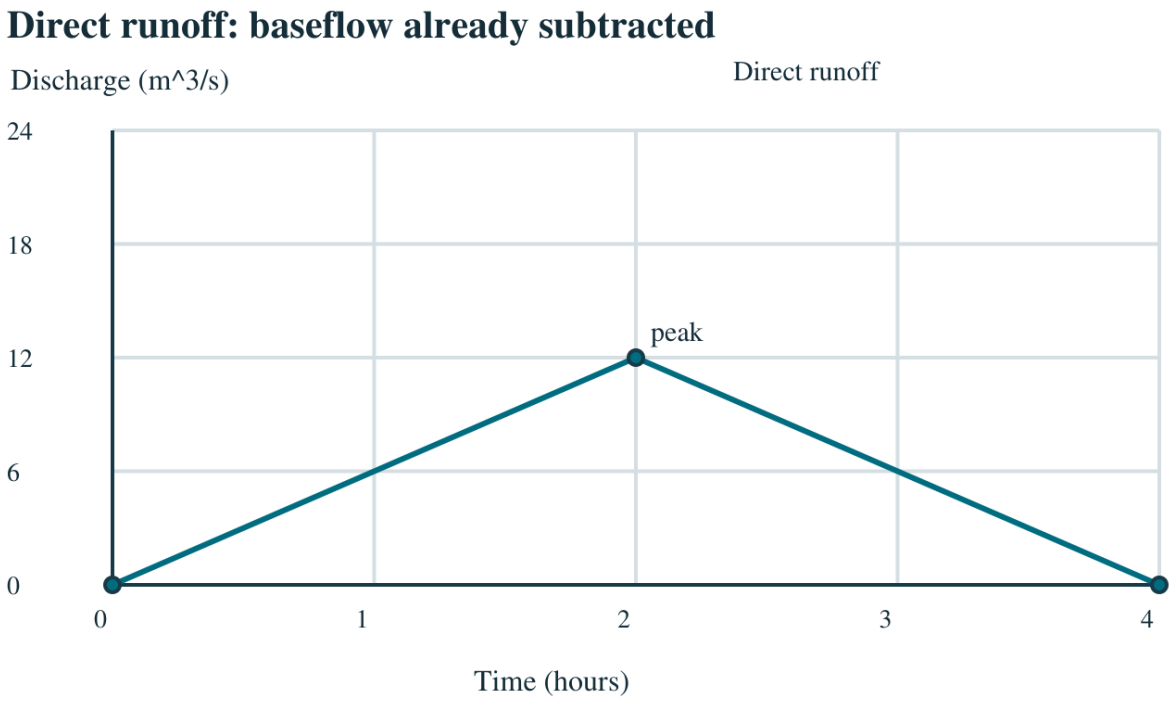
Numerical target 10.8; absolute tolerance 0.0324. Accepted units: , m^3/s.

23. (2 points) Use Figure F1. The water rises while width and mean velocity remain fixed. Explain the predicted discharge change.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

answer: Greater depth increases cross-sectional area, so Q=Av increases under those assumptions. (2 points).

24. (2 points) Use Figure F2. Read the time of peak direct runoff.



F2. Storm hydrograph (original synthetic teaching figure)

answer: 2 hours (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: , hours.

25. (2 points) Use Figure F2. Find the event runoff volume. Convert hours to seconds.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

answer: 86400 m^3 (2 points).

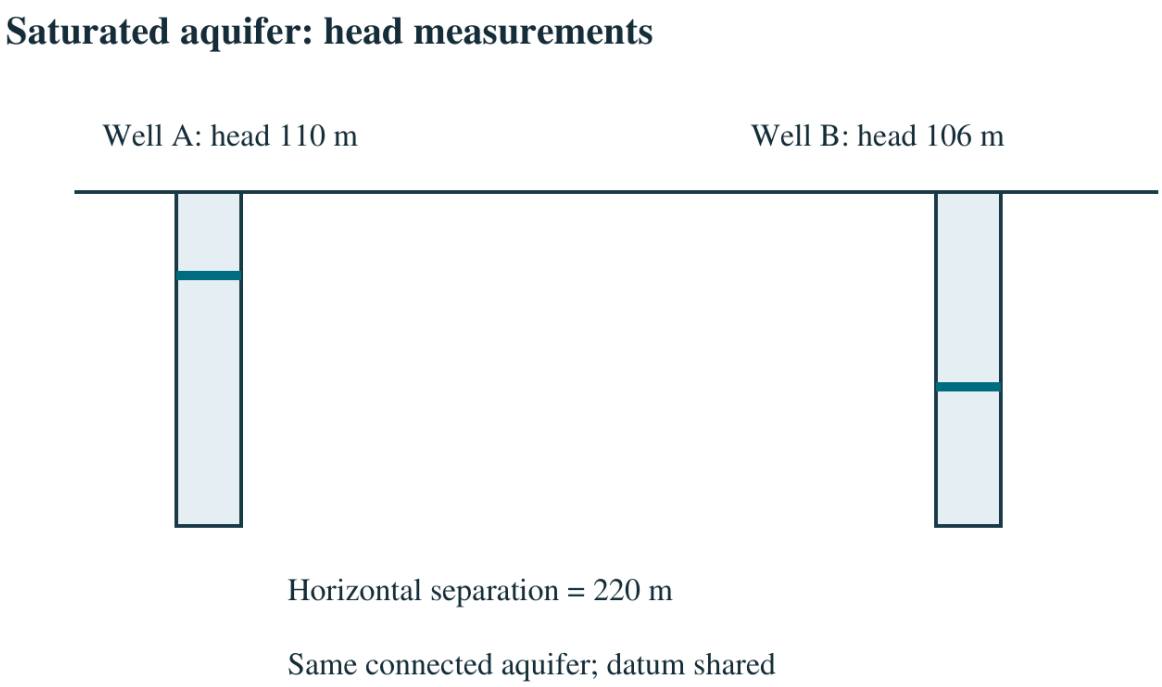
Numerical target 86400; absolute tolerance 259.2. Accepted units: , m^3.

26. (2 points) Use Figure F2. A wetland spreads the same runoff volume over more time. Describe how its hydrograph could differ.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

answer: Its peak can be lower and its duration longer while the area under the discharge-time curve remains the same. (2 points).

27. (2 points) Use Figure F3. Calculate hydraulic-gradient magnitude.



F3. Groundwater heads in two wells (original synthetic teaching figure)

answer: 0.018181818 (2 points).

Numerical target 0.018181818; absolute tolerance 5.4545455e-05. Accepted units: .

28. (2 points) Use Figure F3. Predict the horizontal flow direction and identify the controlling measurement.

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

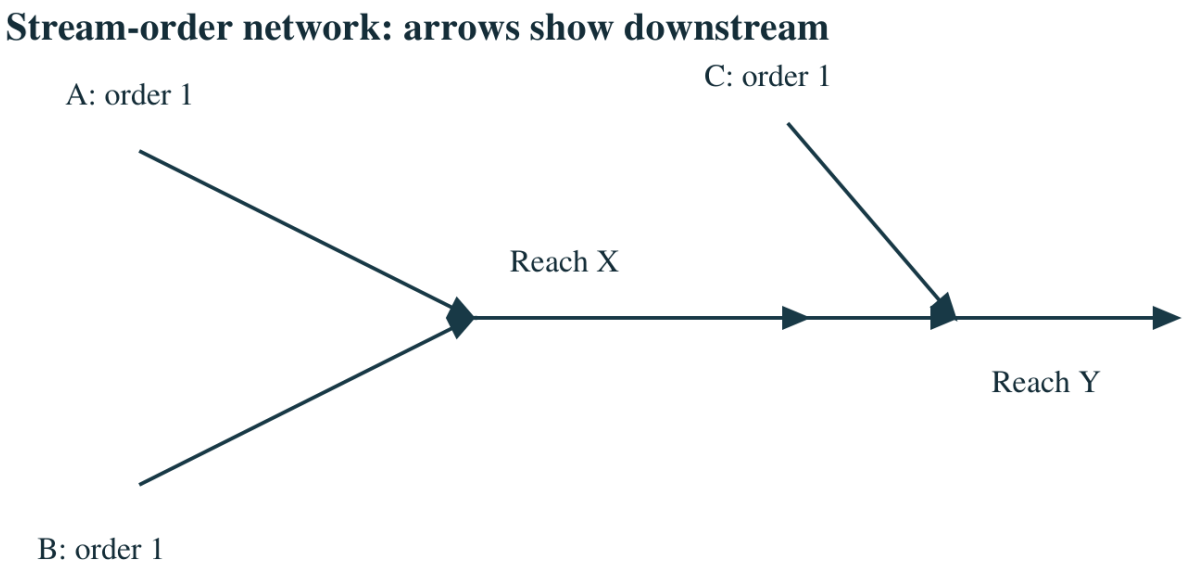
answer: From A toward B, from higher hydraulic head to lower head, assuming the connected aquifer shown. (2 points).

29. (2 points) Use Figure F3. Why cannot the diagram alone determine groundwater discharge?

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

answer: Hydraulic conductivity and flow cross-sectional area are also required for Darcy discharge; head gradient alone is insufficient. (2 points).

30. (2 points) Use Figure F4. Give the Strahler order of reach X.



F4. Tributary network (original synthetic teaching figure)

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

31. (2 points) Use Figure F4. Give the Strahler order of reach Y.

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

32. (2 points) Use Figure F4. If C were second order instead, how would Y change?

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

answer: Two second-order reaches joining would produce a third-order Y. (2 points).